MIL-P-46338B(MU)
31 December 1971
SUPERSEDING
MIL-P-46338A(MU)
25 May 1970

MILITARY SPECIFICATION

PERISCOPE, TANK: M32

1. SCOPE

1.1 This specification covers one type of combination infrared-daylight tank gunner's periscope designated Periscope, Tank: M32. The periscope is a part of the fire control system in the 105MM Gun Tank: M60 and is used for day and night sighting and observing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-B-50366	Body (Daylight) Assembly: 10516210
MIL-H-50368	Head Assembly: 10514220
MIL-B-14843	Body (Infrared) Assembly: 10541950

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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DRAWINGS

U.S. Army Frankford Arsenal

F8247391	Test Fixture for Periscope M32
F8575736	Vibration Adapter for Periscope M32
F10514284	Periscope, Tank: M32
F10540499	Universal Test Stand and Set-Up Instructions
F10547434	10" Collimator
F11732155	Gage, Interchangeability, Minimum
F11732156	Gage, Interchangeability, Maximum

PACKAGING DATA SHEET

10514284	Packaging of Periscope, Tank: M32

(Copies of specifications, standards, drawings and packaging data sheet required by suppliers in connection with specific procurements functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The periscope shall be manufactured in accordance with Drawing F10514284 and drawings pertaining thereto. The following shall have met their respective specifications (except that they need not be subjected to separate vibration tests) and be interchangeable within the requirements of this specification.

Body (Infrared) Assembly:	10541950	MIL-B-14843
Body (Daylight) Assembly:	10516210	MIL-B-50366
Head Assembly :	10514220	MIL-H-50368

3.2.1 Interchangeability.

a. Modules.- The periscope shall meet the requirements of 3.6.2, 3.6.5, 3.6.6 and 3.6.8 without need for realignment of components or selection of modules listed in 3.2. Adjustments, other than those specified herein, are prohibited.

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b. Periscope with mount.- The periscope shall be capable of insertion into the gages specified in 4.6.5.2. After insertion, the periscope shall be capable of being firmly secured and subsequently removed.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Vibration.- This requirement shall be met with the image converter tube and reticle projector lamp energized by the application of +24 ± 6.0 vdc to the contract for the IR power supply and the reticle projector lamp (Referenct Drawing F10541950). The periscope shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/6 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. There shall be no failure or flicker of light emission from the image converter tube and reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test the periscope shall meet all remaining requirements and then each module (see 3.2) shall meet their own respective post vibration test requirements.

3.5 Cleanliness.- With the periscope disassembled, the three modular units shall meet the following requirements:

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates fractures, and adhesive separations on any glass component when viewing through either end of each module.

b. Dirt.

(1) Body assembly (daylight).- When viewed through the eyepiece, there shall be no more than three dirt particles appearing within a 50 mil diameter central field of view. There shall be no more than three dirt particles appearing outside the specified field of view. The angular separation between dirt particles shall be at least 15 mils. Dirty particles smaller than 1/4 reticle line width shall be ignored regardless of distribution. The size of any particle shall not exceed the apparent width of the reticle line.

(2) Body assembly (infrared). When viewing through the eyepiece with the image converter tube energized, there shall be no more than five particles visible on the screen surface. The size of any particle shall not exceed the apparent width of the reticle line.

(3) Head assembly.- There shall be no more than three particles of dirt or foreign material (clased as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05 inch.

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3.6 Performance.

3.6.1 Orientation.

a. Datum Plane "W" defined by the four mounting pads in a vertical plane. Datum Plane "Y" on a horizontal plane.

b. Side "L" of the input coupling positioned 22 degrees 30 minutes with respect to Datum Plane "W" as measured in a vertical plane.

3.6.2 Parallax.- Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.1 mil when adjusted by Eccentric 8205635 on the daylight body assembly.

3.6.3 Collimation.- The following collimation requirements shall be established prior to meeting the vibration requirements of 3.4:

a. An incident line of sight which is originally perpendicular within 0.2 mil to Datum Plane "W".

b. The input coupling positioned in accordance with 3.6.1b.

3.6.3.1 Pre-vibration setting.- Rotate visible and infrared boresight knobs to establish coincidence of each boresight cross with the incoming line of sight (see 3.6.3a) prior to meeting the requirements of 3.4.

3.6.3.1.1 Daylight viewing area:

a. Elevation knob travel.- The boresight knob shall provide a minimum travel of 5.0 mils up and 8.0 mils down from the collimated setting of 3.6.3a.

b. Deflection knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.6.3a.

c. Line of sight variation.- Variation in the line of sight in elevation as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.6.3a shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.6.3.1.2 Infrared viewing area:

a. Elevation knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in elevation and depression from the collimated setting of 3.6.3a.

b. Deflection knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.6.3a.

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c. Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.6.3a shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.6.3.2 Post vibration collimation.- The original setting of each reticle boresight relative to the incoming line of sight established in 3.6.3.1 shall not deviate more than 0.2 mil (including fixture error) as observed through eyepiece subsequent to meeting requirements of 3.4.

3.6.4 Collimation repeatability.

3.6.4.1 Assembling daylight and infrared bodies to head assembly.- Each body assembly shall be capable of repeating an original line of sight within 0.20 mil when removed and reinstalled (a minimum of 3 times) into the head assembly.

3.6.5 Image tilt - daylight & infrared.- The image of a target line which is originally parallel to Datum Plane "Y" shall emerge from the eyepiece parallel to this same surface within 2.0 degrees of arc.

3.6.6 Parallelism of reticle & image, daylight & infrared.- The image of a target line originally parallel to Datum Plane "Y" shall be parallel to the horizontal reticle line within 2.0 degrees of arc.

3.6.7 Objective focusing range.

3.6.7.1 Daylight.- Rotation of Eccentric 8205635 shall cause the objective assembly to travel a minimum of 0.010 inch in each direction from the position established in 3.6.2.

3.6.7.2 Infrared.- Rotation of Ring 8620059 shall cause the objective to bring targets from 200 meters to 1200 meters into focus.

3.6.8 Resolution.

3.6.8.1 Daylight.- The resolution shall be 8.0 seconds of arc or less in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.6.8.2 Infrared.- The resolution shall be 1 minute of arc or less in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.6.9 Electrical characteristic - infrared.

3.6.9.1 Image converter tube.- The image converter tube shall be energized by the application of +24 ±6.0 vdc to the contact for the IR Power Supply (Ref. Dwg F10541950).

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3.6.9.2 Image steadiness.- There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.6.9.3 Reticle illumination.- With a minimum of 18 and a maximum of 30 volts applied to the contact for the reticle lamp (Ref Dwg F10541950) the reticle marking shall be visible when viewing through the eyepiece with the image tube energized (see 3.6.9.1).

3.6.10 Operability, focusing lock.- This requirement shall be met with a 40 ounce inches (oz-in.) torque applied to the locking screw on Housing 8634490. There shall be no rotation of Eccentric 8205635 when 48 oz-in torque is applied to the eccentric.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three periscopes plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for

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materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP listed in the technical data package (TDP)). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Line of sight variation, daylight viewing area	3.6.3.1.1c	4.6.4.3.3
2.	Line of sight variation, infrared viewing area	3.6.3.1.2c	4.6.4.4.3
3.	Resolution, daylight viewing area	3.6.8.1	4.6.4.8
4.	Resolution, infrared viewing area	3.6.8.2	4.6.4.9
5.	Image converter tube	3.6.9.1	4.6.4.1.1
6.	Image steadiness	3.6.9.2	4.6.4.1.2
7.	Reticle illumination	3.6.9.3	4.6.4.1.3
8.	Interchangeability, periscope with mount	3.2.1b	4.6.5.2

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between +60° and +90°F.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS CRITICAL: NONE MAJOR: AQL 0.65% Defective	REQUIREMENT	TEST PROCEDURE
101. Pre-vibration collimation setting	3.6.3.1	4.6.1
102. Vibration (periscope)	3.4. .	4.6.2
103. Post-vibration collimation	3.6.3.2	4.6.2.1 & 4.6.2.1.1
104. Cleanliness	3.5	4.6.3
105. Parallax	3.6.2	4.6.4.2 & 4.6.4.2.1
106. Elevation knob travel, daylight viewing area	3.6.3.1.1a	4.6.4.3.1
107. Deflection knob travel, daylight viewing area	3.6.3.1.1b	4.6.4.3.2
108. Elevation knob travel, infrared viewing area	3.6.3.1.2a	4.6.4.4.1
109. Deflection knob travel, infrared viewing area	3.6.3.1.2b	4.6.4.4.2
110. Collimation repeatability, assembling daylight body to head assembly	3.6.4.1	4.6.2.1.2.1
111. Collimation repeatability, assembling infrared body to head assembly	3.6.4.1	4.6.2.1.2.2
112. Image tilt, daylight and infrared	3.6.5	4.6.4.5
113. Parallelism of reticle and image, daylight and infrared	3.6.6	4.6.4.6
114. Objective focusing range, daylight	3.6.7.1	4.6.4.7.1
115. Objective focusing range, infrared	3.6.7.2	4.6.4.7.2
116. Operability, focusing lock	3.6.10	4.6.6
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable drawings and specifications - visual
302.	General specification	3.3	MIL-F-13926 - visual

4.4.1.1 Three periscopes that have met the requirements and tests in 4.3.2.2 and in Table I shall be selected at random as a special sample from each months production. Each sample shall then meet the requirement and test in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Interchangeability, modules	3.2.1a	4.6.5.1

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered

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to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

4.5.2.3.1 Performance.- Special test equipment to determine performance of the periscope shall conform to the requirements of Drawing F8247391 (Test Fixture for Periscope, M32) and Drawing F10540499 (Universal Test Stand and Set Up Instructions). The "Test Components" depicted on Drawing F10540499 form a part of the test equipment.

4.6 Test methods and procedures.

NOTE: At no time during the following tests shall the input coupling be allowed to strike hard against the travel stops.

4.6.1 Pre-vibration collimation setting.- This test and the following test of 4.6.2.1.1 shall be performed with the periscope positioned and adjusted as in 4.6.4. Prior to performing these tests, disengage the index pin "D" from the index plate of the test fixture and place the 10 second level on surface "C" of the leveling base. The bubble of the level shall be centered within its graduation lines. If it is not, then adjustment is required in accordance with the procedures prescribed in paragraphs 7 and 8 of the set-up instructions contained on Drawing F8247391. With the index pin "D" disengaged from the index plate and surface "C" of the leveling base level, view through each periscope eyepiece (daylight and infrared) and obtain coincidence of each periscope reticle boresight cross respectively, and the projected target image crosslines by means of the daylight and infrared boresight knobs (do not slip knob scales). When coincidence has been obtained, record the boresight knob scale settings for use in the test in 4.6.2.1. Without disturbing the knob scale settings, the periscope shall be then subjected to the vibration test in 4.6.2.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original recorded settings throughout remainder of tests herein.

4.6.2 Vibration (periscope).- Inspection equipment capable of providing the amplitude and frequency specified in 3.4 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The periscope shall be clamped on the vibrating equipment by means of an adapter that conforms to Drawing F8575736, Vibration Adapter for Periscope M32. The vibration motion shall be in the vertical plane. During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Prior to vibration, energize the image converter tube and reticle projector lamp as specified in 3.4. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (infrared). The periscope shall then be vibrated

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for the time duration specified in 3.4. During the first, third and fifth minutes of the vibration cycle, observe the screen surface of the tube and the reticle pattern. During these observations, there shall be no failure or flicker of light emission from the image converter tube and the reticle projector lamp.

NOTE: When provided as part of a complete periscope, vibration of each head and body assembly, including their respective post-vibration collimation test, will not be required.

4.6.2.1 Post-vibration.- Immediately following vibration, the periscope shall again be positioned and adjusted as in 4.6.1 and then subjected to the post-vibration collimation test of 4.6.2.1.1. Insure that the boresight knob scales are not disturbed from the original setting obtained in 4.6.1.

4.6.2.1.1 Post-vibration collimation.- Observe through each eyepiece assembly (daylight and infrared) and adjust by means of the boresight knobs to obtain coincidence of each periscope reticle boresight cross with the target image lines. Boresight knob adjustment shall not exceed the tolerance specified in 3.6.3.2, as measured on the boresight knob scales. Record each collimation setting for use in the tests in 4.6.2.1.2.

4.6.2.1.2 Collimation repeatability.

4.6.2.1.2.1 Assembling daylight body to head assembly.- Perform this test subsequent to test in 4.6.2.1.1. Remove the daylight body from the head assembly, removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the daylight collimation setting, recorded in 4.6.2.1.1. The periscope shall meet the requirements specified in 3.6.4.1, as measured on the boresight knobs.

4.6.2.1.2.2 Assembling infrared body to head assembly.- Perform this test subsequent to the test in 4.6.2.1.2.1. Remove the infrared body from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the infrared collimation setting, recorded in 4.6.2.1.1. The periscope shall meet the requirements specified in 3.6.4.1, as measured on the boresight knobs.

4.6.2.1.3 Post-vibration examination.- The periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4, prior to being subjected to the remaining tests specified herein. A visual examination shall also be made for missing or damaged parts, glass chips foreing matter and bonding (separations, chipping or rupture). An audible examination shall also be made of the periscope for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a periscope defective.

4.6.3 Cleanliness.- Subsequent to the post-vibration tests, each separate assembly (see 3.2) shall be subjected to the tests in 4.6.3.1, 4.6.3.2 and 4.6.3.3 respectively to determine compliance with 3.5.

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4.6.3.1 Body assembly (daylight).- The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly (daylight) shall conform to the requirements of 3.5a and 3.5b(1).

4.6.3.2 Body assembly (infrared).- The image converter tube shall be de-energized for this portion of the test. With 2 each 15 watt desk fluorescent lamps held close to the objective window and eyelens, view by reflected light the glass components of the body assembly to determine compliance with the cleanliness requirements of 3.5a. Upon successful completion of this portion of the test, energize the image converter tube from a power source capable of supply and maintaining 24 volts DC. The image tube shall be permitted a stabilization period of 15 seconds prior to performing the cleanliness test. Subsequent to the tube stabilization period, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter ring. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.5b(2) and shall conform to the requirements specified therein. "Cold" and "Hot" spots on the screen surface of the image tube shall not be considered dirt.

4.6.3.3 Head assembly.- The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the exit window, inspection for dirt shall be made by viewing the illuminated field specified in 4.6.3.1. The head assembly shall conform to the requirements of 3.5a and 3.5b(3).

4.6.4 Performance.- To determine compliance with 3.6, the following tests shall be performed with the complete periscope positioned on the special test equipment specified in 4.5.2.3.1. Set-up of the test equipment and installation of the periscope under test shall be performed in accordance with the set-up and calibration instructions contained on Drawing F8247391. Unless otherwise specified during these tests, the Infrared Focusing Ring 8620059 shall be adjusted for best focus of the target image after the diopter adjustment knob of the infrared eyepiece is set for best focus of the phosphor grain on the screen of the image tube; Eccentric 8205635 (for adjusting parallax) shall be adjusted for minimum parallax between the target image and reticle and then locked after the diopter adjustment knob of the daylight eyepiece is set for best focus of the periscope reticle: the Collimator Projector 10547434 shall be focused at 1200 meters.

4.6.4.1 Electrical characteristics - infrared.

4.6.4.1.1 Image converter tube.- Apply the voltage specified in 3.6.9.1 to the contact for the IR power supply (reference Drawing F10541950) and position the electrical switch to the "24 volt" position. After allowing a 15 second tube stabilization period, view through the infrared eyepiece and

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observe the image tube screen. The screen surface of the tube shall be illuminated.

4.6.4.1.2 Image steadiness.- Perform this test with the reticle projector lamp off. Position an observation telescope at the infrared eyepiece of the periscope. The telescope shall contain a crossline reticle pattern and be capable of focusing on the screen of the image tube. With the aid of the telescope, view through the infrared eyepiece and establish the telescope reticle crosslines coincident to the projected target image crosslines. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker, no movement of the image nor audible sparking as specified in 3.6.9.2. Image movement shall be determined by means of the target image with respect to the observation telescope reticle pattern. Sparking shall be determined by means of an audio examination. Upon completion of this test, energize the reticle projector lamp.

4.6.4.1.3 Reticle illumination.- Perform this test concurrent with the test in 4.6.4.1.2 and while the reticle projector lamp is energized at each voltage specified in 3.6.9.3. The reticle markings shall appear clearly defined at each required voltage setting when visually inspected through the infrared eyepiece.

4.6.4.2 Parallax.- View through the daylight eyepiece and establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While viewing through the eyepiece, parallax will be recognized by any apparent displacement of the intersecting point of the periscope reticle boresight cross with respect to the intersecting point of the projected target image crosslines when the observer's head is moved from side to side or up and down. Displacement between the periscope reticle and target image shall not exceed the tolerance specified in 3.6.2. Upon satisfactory completion of this test, perform the tests specified in 4.6.4.2.1 and 4.6.6.

4.6.4.2.1 Parallax adjustment.- If adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw on Housing 8634490 and Eccentric 8205635 shall be utilized for performing this adjustment. Back off the locking screw a minimum of two revolutions from its locked position. Then while performing the test in 4.6.4.2, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.6.10 to the locking screw utilizing the torque wrench specified in 4.6.6.

4.6.4.3 Daylight viewing area.

4.6.4.3.1 Elevation knob travel.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.6.3.1.1a as measured on the target. Re-establish coincidence of the reticle boresight cross

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and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall travel as specified in 3.6.3.1.1a as measured on the target.

4.6.4.3.2 Deflection knob travel.- Perform this test utilizing the procedure in 4.6.4.3.1 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.6.3.1.1b as measured on the target.

4.6.4.3.3 Line of sight variation.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.6.3.1.1c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.6.3.1.1c to determine overall conformance with the requirements specified therein.

4.6.4.4 Infrared viewing area.

4.6.4.4.1 Elevation knob travel.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a as measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a as measured on the target.

4.6.4.4.2 Deflection knob travel.- Perform this test utilizing the procedure in 4.6.4.4.1 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.6.3.1.2b as measured on the target.

4.6.4.4.3 Line of sight variation.- Establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob, any variation in

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elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.6.3.1.2c. Re-establish coincidence of the periscope reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotate the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the values specified in 3.6.3.1.2c to determine overall conformance with the requirements specified therein.

4.6.4.5 Image tilt - daylight and infrared.- The daylight and infrared optical systems of the periscope shall be inspected utilizing an auxiliary telescope containing a crossline reticle pattern. The vertical reticle line of the auxiliary telescope shall be aligned plumb when in the test position on a qualified level surface. With aid of the auxiliary telescope, view through the infrared eyepiece of the periscope and establish one extremity of the telescope vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.6.5, as measured on the target. Repeat the above procedure with the auxiliary telescope positioned at the daylight eyepiece of the periscope. The images formed by the two optical systems (daylight and infrared) of the periscope shall meet the requirements specified in 3.6.5.

4.6.4.6 Parallelism of reticle and image - daylight and infrared.- Perform this test while viewing through the infrared eyepiece of the periscope with the unaided eye. Position one extremity of the periscope horizontal reticle line coincident to the horizontal target image line. The periscope horizontal reticle line shall be parallel to the horizontal target line within the tolerance specified in 3.6.6, as measured on the target. The above procedure shall also be repeated at the daylight eyepiece of the periscope in compliance with 3.6.6.

4.6.4.7 Objective focusing range.

4.6.4.7.1 Daylight.- This test shall be performed subsequent to the test in 4.6.4.2. A dial indicator, graduated in .001 of an inch, mounted on an appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested and the screw driver specified in 4.6.4.2.1 shall be utilized for this test. With the daylight body removed from the periscope and firmly secured on the supporting medium, position the dial indicator so that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero with the indicating hand. Back off the locking screw of Housing 8634490 a minimum of two full revolutions from its locked position. Starting from the zero reading of the dial indicator, rotate the eccentric until the objective assembly has traveled the minimum distance specified in 3.6.7.1,as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this

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travel in the opposite direction. There shall be sufficient travel in each direction to meet the minimum requirements specified in 3.6.7.1, as measured on the dial indicator. Upon completion of this test, reinstall the daylight body in periscope and repeat the parallax adjustment procedure specified in 4.6.4.2.1. The daylight body shall then be subjected to the test in 4.6.6.

4.6.4.7.2 Objective focusing range - infrared.- The test for short range focus requirements of the infrared optical system of the periscope shall be performed with the target image adjusted for 200 meters. With the diopter ring positioned for best focus of the phosphor grain on the screen of the image tube, rotate infrared focusing ring of the periscope to short range setting. The periscope shall conform to the requirements specified in 3.6.7.2. The test for long range focus requirements of the periscope shall be performed with the target image adjusted for 1200 meters. Rotate the infrared focusing ring to the long range setting to determine compliance with the requirements specified in 3.6.7.2.

4.6.4.8 Resolution - daylight.- The daylight optical system of the periscope shall be inspected for resolution utilizing a calibrated dioptometer with a mangification of at least 3 power (3X). Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter adjustment knob of the periscope daylight eyepiece to indicate zero diopter. Position the dioptometer at the periscope daylight eyepiece. While viewing through the dioptometer, adjust the objective scale of the dioptometer for best focus of the resolution target image. Any change in diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.6.8.1.

4.6.4.9 Resolution - infrared.- The test for resolution shall be performed utilizing a supporting medium capable of receiving the periscope and a resolving power chart or a wall target representing the angular subtense for minutes of arc specified in 3.6.8.2. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and contain four lines sets as shown on Figure I therein. The target shall be illuminated by means of a filtered tungsten lamp. The filter shall not pass visible light rays below 0.65 microns. Place the periscope on the supporting medium and position it on its side with the input coupling arm positioned in accordance with 3.6.3b. Energize the image tube of the periscope from a power source capable of suppling and maintaining 24 volts DC. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting the resolution test of the infrared optical system. Upon completion of the stabilization period, view through the infrared eyepiece with the unaided eye and obtain best focus of the numeral in the center of the resolving power chart by means of the infrared eyepiece diopter adjustment knob scale and Ring 8620059. Any change in the diopter readings by adjustment of the infrared eyepiece diopter knob scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the knob scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.6.8.2.

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4.6.5 Interchangeability.

4.6.5.1 Modules.- Each sample periscope (see 4.4.1.1) shall be disassembled (removal of infrared body from daylight body and the daylight body from head) to permit interchange of the component modules. The modules shall be identified and interchanged as shown in Table IV.

TABLE IV

Original Module Positions

	Periscope 1	Periscope 2	Periscope 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-1	DL-2	DL-3
Body Assembly (IR) (Infrared)	IR-1	IR-2	IR-3

Interchanged Module Positions

	Periscope 1	Periscope 2	Periscope 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-2	DL-3	DL-1
Body Assembly (IR) (Infrared)	IR-3	IR-1	IR-2

Upon reassembly of the three interchanged periscopes, each periscope shall be subjected to the tests specified in 4.6.4.2, 4.6.4.5, 4.6.4.6, 4.6.4.8 and 4.6.4.9 to determine compliance with 3.2.1a.

4.6.5.2 Periscope with mount.- This test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11732156 and F11732155 respectively, to determine compliance with 3.2.1b.

4.6.6 Operability, focusing lock.- This test shall be performed in sequence to the test in 4.6.4.7.1. A dial indicator, graduated in increments of .0001 of an inch, mounted on an appropriate stand, the supporting medium specified in 4.6.4.7.1 and a torque wrench capable of applying the specified torque values in 3.6.10, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. Remove the daylight body from the periscope and secure firmly on the supporting medium. With the specified torque applied to the locking screw, position the dial indicator so

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that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero. Apply, in a clockwise direction, the specified torque to the eccentric, then observe the dial indicator reading. Then apply, in a counterclockwise direction, the specified torque to the eccentric and again observe the dial indicator reading. To conform to the requirements of 3.6.10, there shall be no change in the established dial indicator zero reading when observed after each torque application. Upon completion of this test, reinstall the daylight body in the periscope.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10514284. The level of procurement shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This periscope is utilized as a day or night sighting and observation instrument. This capability is made possible by the use of a infrared sighting system and a conventional periscope optical system encased in a single body assembly. The infrared portion utilizes the 24 volt vehicle current source which is converted to a negative 16,000 volts, the required voltage to actuate the image converter tube.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable package data sheet (see 5.1)
d. Applicable stock number
e. Provisions for First Article Testing

6.3 Input coupling.- At no time shall the input coupling be allowed to strike hard against the travel stops.

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